Customer Churn Prediction App

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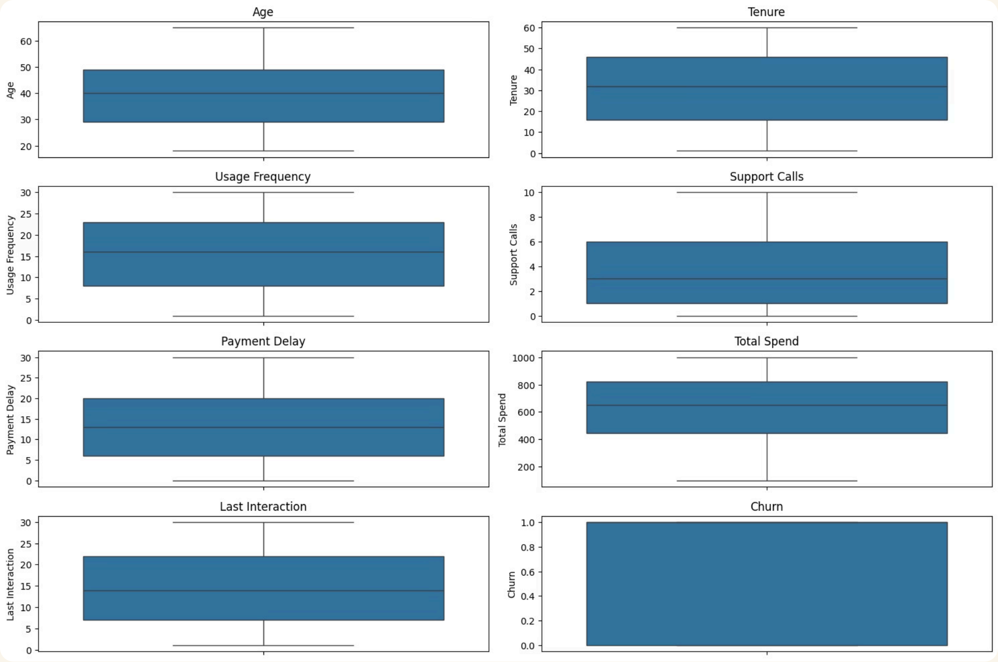
Concise technical briefing for the data team and supervisors: focus on cleaning, EDA highlights, feature engineering, model comparison and final deployment choice. Visuals emphasise the exact figures/plots used during analysis.

Data loading, merge & cleaning — quick summary

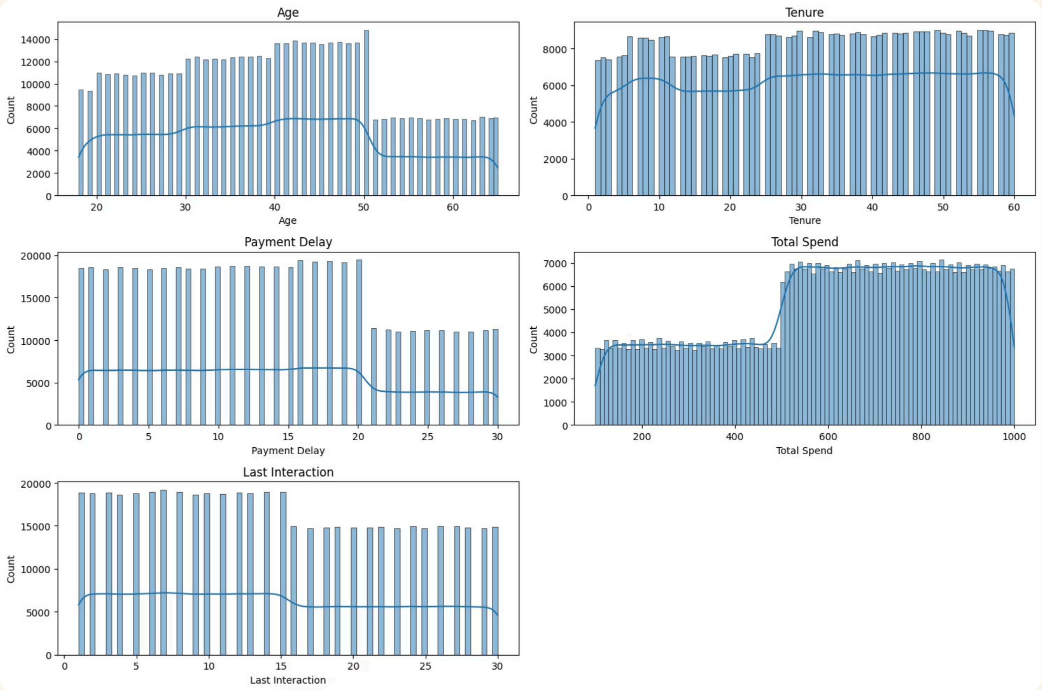
- **Data Consolidation & Cleaning:** Merged train and test sets (505K+ rows), dropped `CustomerID`, removed a single null row, and confirmed zero duplicates.
- **Integrity Validation:** Verified all numerical features are strictly positive (no logical errors or extreme outliers), confirmed clean categorical labels (no typos), and identified a manageable class imbalance.

Transition Note: Statistical descriptions suggest a clean distribution with **no extreme outliers**; a hypothesis to be visually confirmed in the Univariate stage.

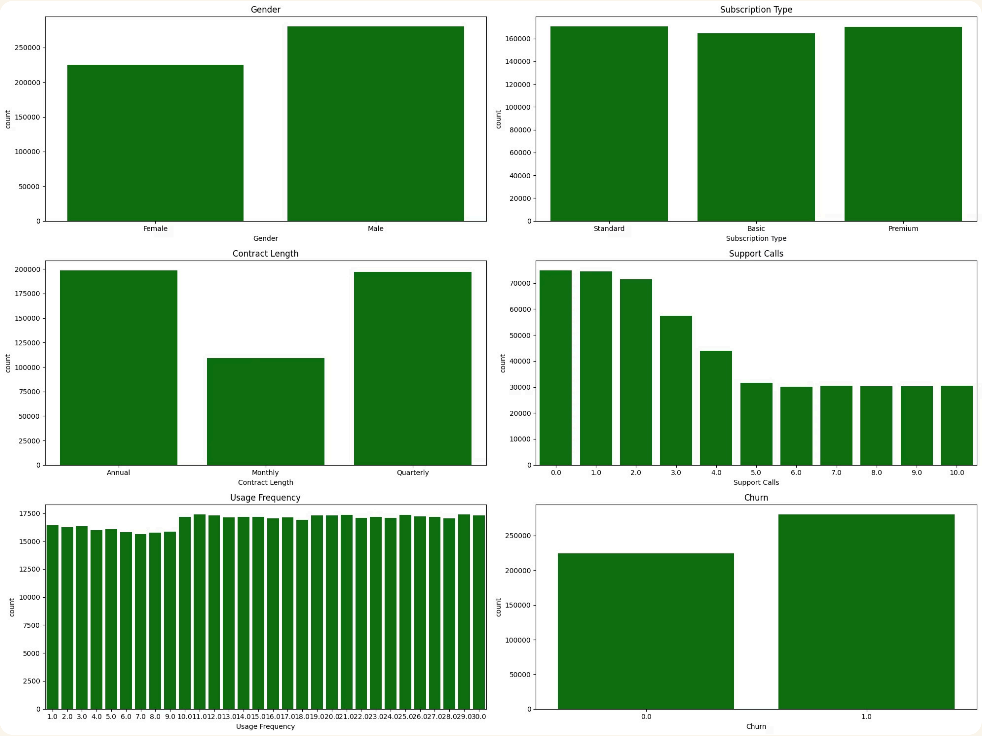
Univariate analysis :



- **Outlier Verification:** Confirmed a complete absence of outliers across all continuous numerical features.
- **Pipeline Impact:** Validated our initial statistical hypothesis, eliminating the need for capping or trimming during preprocessing.

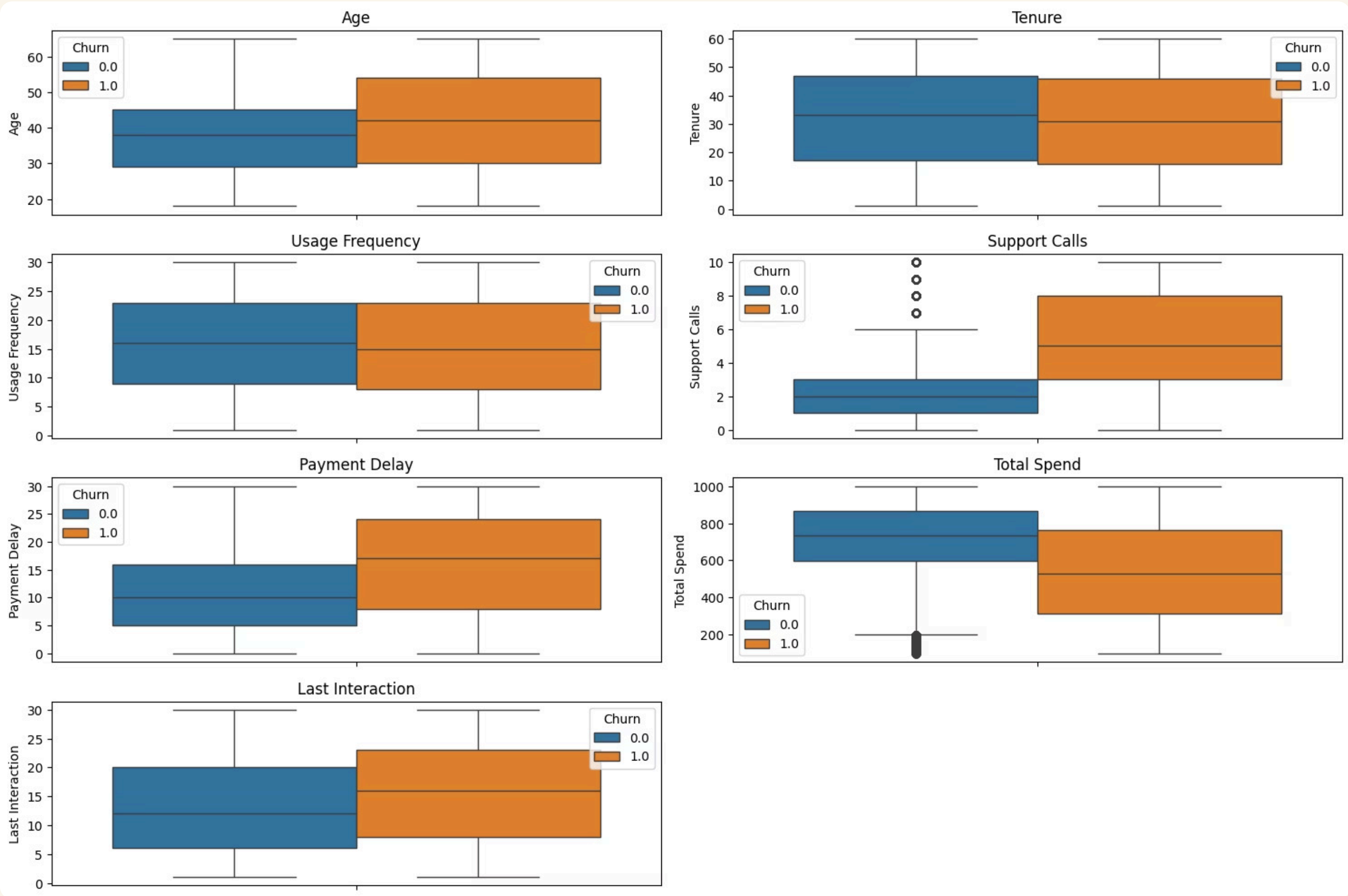


- **Non-Normal Distributions:** Revealed unnatural, sharp frequency drops at specific thresholds (e.g., Age 50, Delay 20).
- **Behavioral Boundaries:** Strongly suggests the presence of rigid data generation rules rather than organic customer variance.hresholds (Age 50, Delay 20, Interaction 15).



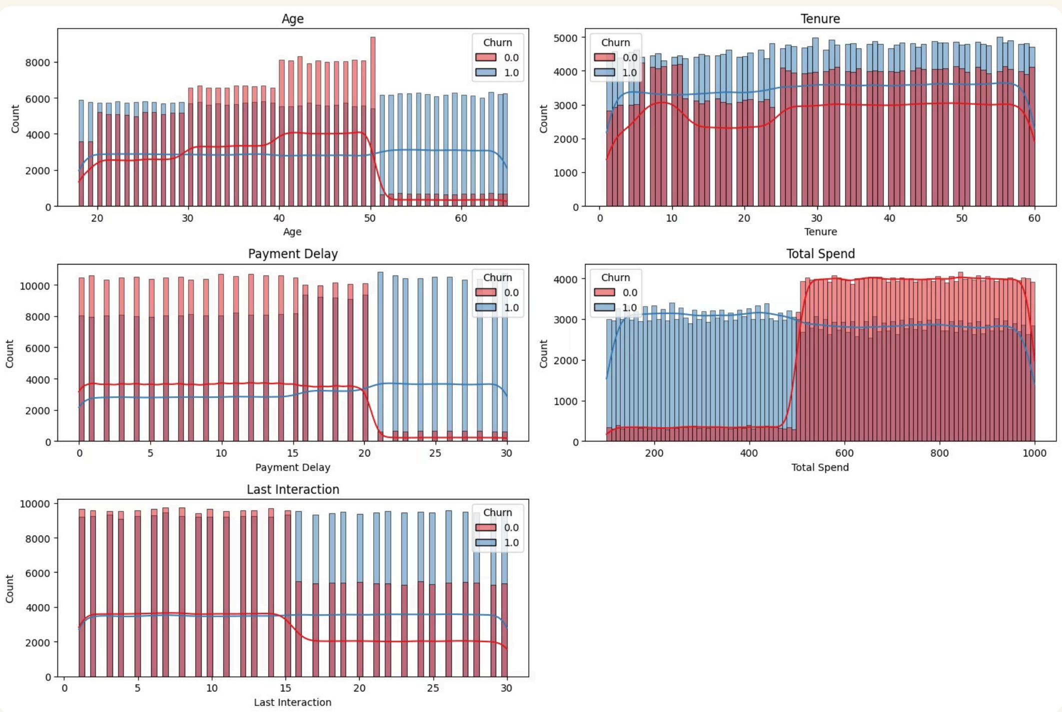
- **Uniformity Anomaly:** Highlighted perfectly flat, uniform distributions across features like Usage Frequency and Subscription Type.
- **Synthetic Confirmation:** Combined with an illogical >50% overall churn rate, this conclusively proves the dataset is artificially generated or oversampled.

Bivariate analysis — numerical vs Churn

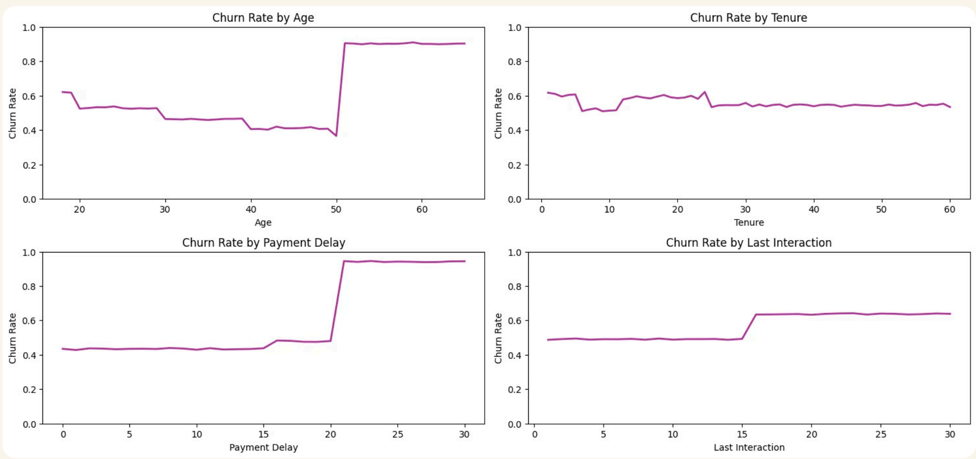


- **Visible Class Separation:** Support Calls and Total Spend show the most distinct separation between churned and retained customers.

- **Hidden Non-Linearity:** Features like Age show overlapping medians, indicating that simple linear models will struggle to separate classes based on these features alone.



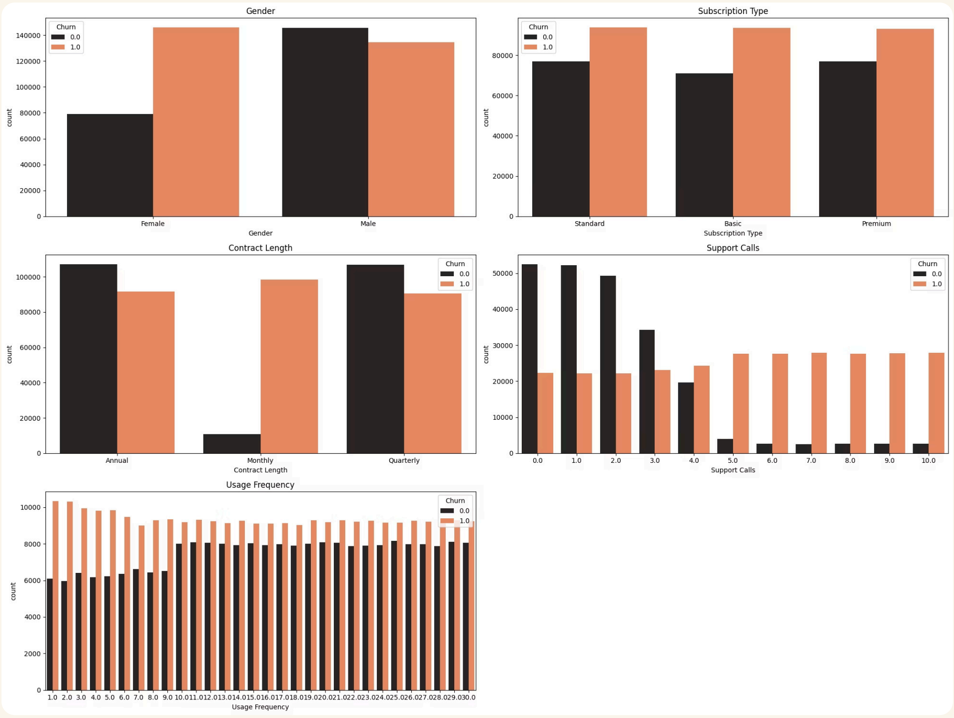
- **Contextual Proportions:** While absolute churn counts seem deceptively flat across ages and spending, comparing them to the base population reveals massive risk disparities.
- **Logical Clustering:** Retained customers are tightly clustered in safe zones (payment delay < 20 days, spend > $500).

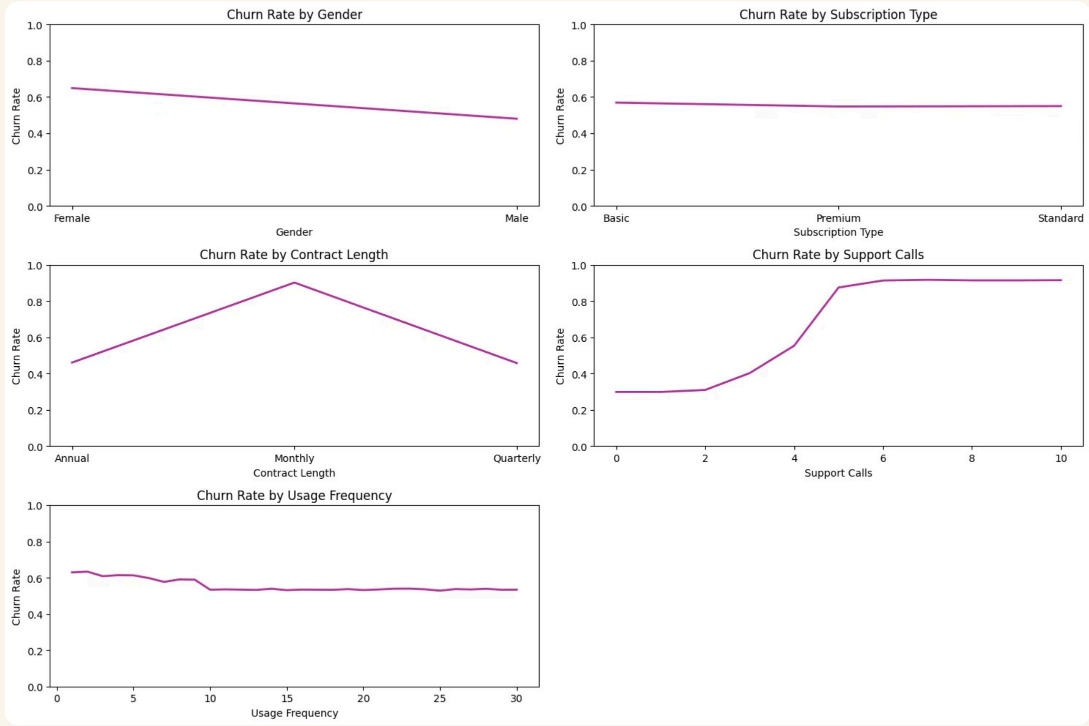


- **Threshold Validation:** Calculating the explicit churn probability eliminates population ambiguity and scientifically proves our hypothesis.

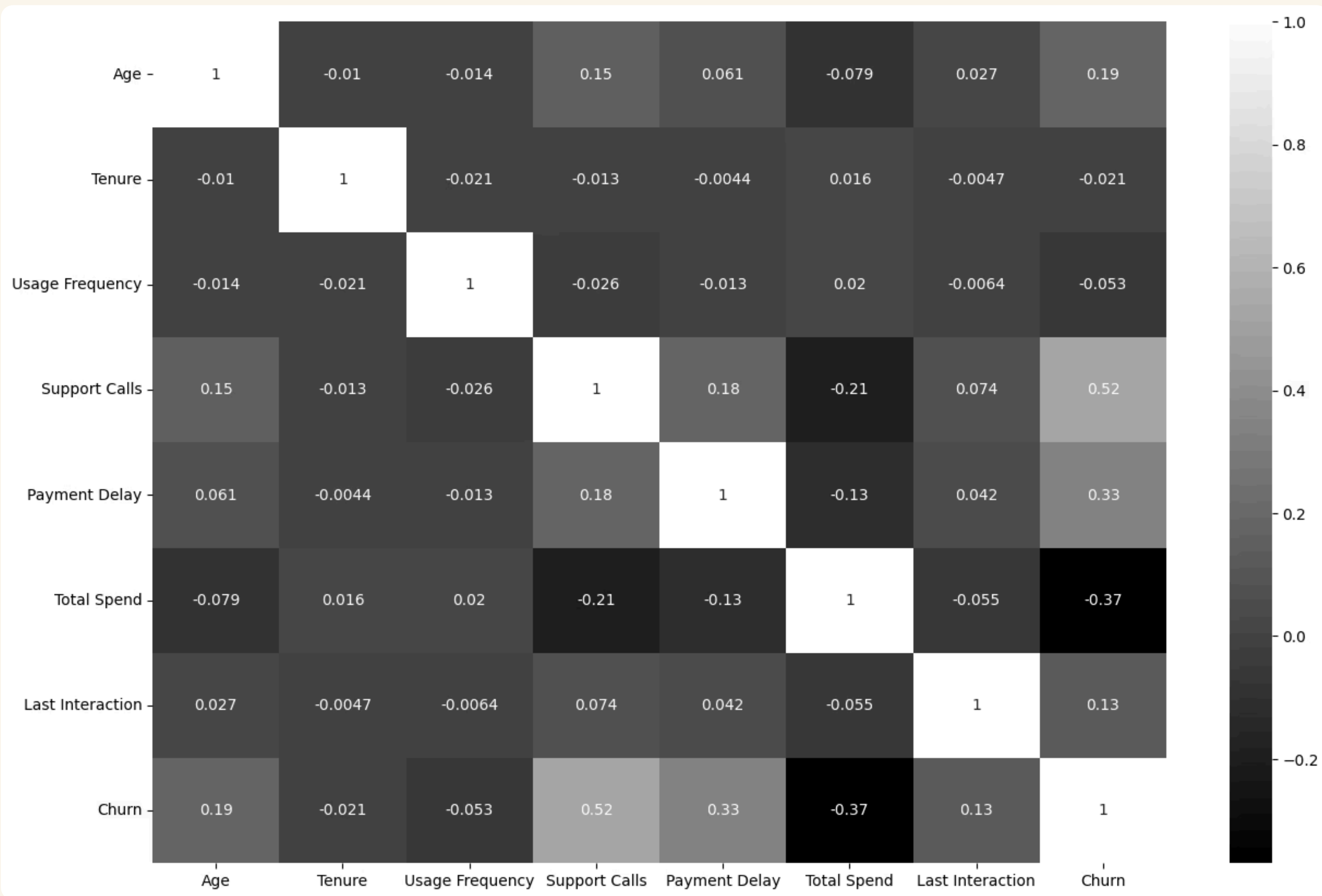
- **Critical Triggers Confirmed:** Validates exact "danger zones" where churn spikes abruptly: Age > 50, Delay > 20 days, and Last Interaction > 15 days.

Categorical features vs Churn





- **Volume Illusion:** Absolute churn counts can be de**Misleading Numbers:** Looking only at raw counts can be tricky. For example, Monthly and Annual contracts seem to have similar churn numbers.
- **Exposing True Risk:** Transforming counts to **High-Risk Groups:** Calculating the actual rates reveals the real danger: Monthly contracts have a huge ~75% churn rate, and female customers churn at ~65%.
- **The Real Picture:** To see the true risk, we must look at percentages (Churn Rate) instead of just total numbers.
- **Weak Predictors:** 'Subscription Type' and 'Usage Frequency' show a flat ~55% churn rate everywhere. They won't help the model much in predicting churn.



- **No Feature Overlap (Multicollinearity):** The numerical features are independent of each other (correlation is almost zero).
- **Great for Modeling:** This is perfect for machine learning because the features won't confuse the model with repeated information.
- **Strongest Triggers:** The highest correlations are strictly between the features and the target (Churn).
- **Confirming our EDA:** Math proves what we saw earlier: Support Calls (+0.52) is the biggest churn driver, and Total Spend (-0.37) is the best reason for staying.

Feature engineering — encoding thresholds into features

Critical_Support_Calls

Binary: 1 if Support Calls > 3 — captures near-deterministic churn behaviour after 4 calls.

Severe_Payment_Delay

Binary: 1 if Payment Delay > 20 days — marks high-risk late payers.

High_Risk_Age

Binary: 1 if Age > 50 — isolates demographic risk segment.

Low_Spend_Flag & Avg_Monthly_Spend

Low_Spend_Flag = Total Spend < 500; Avg_Monthly_Spend = Total Spend / Tenure — normalises monetary behaviour.

Data Preprocessing & Engineering

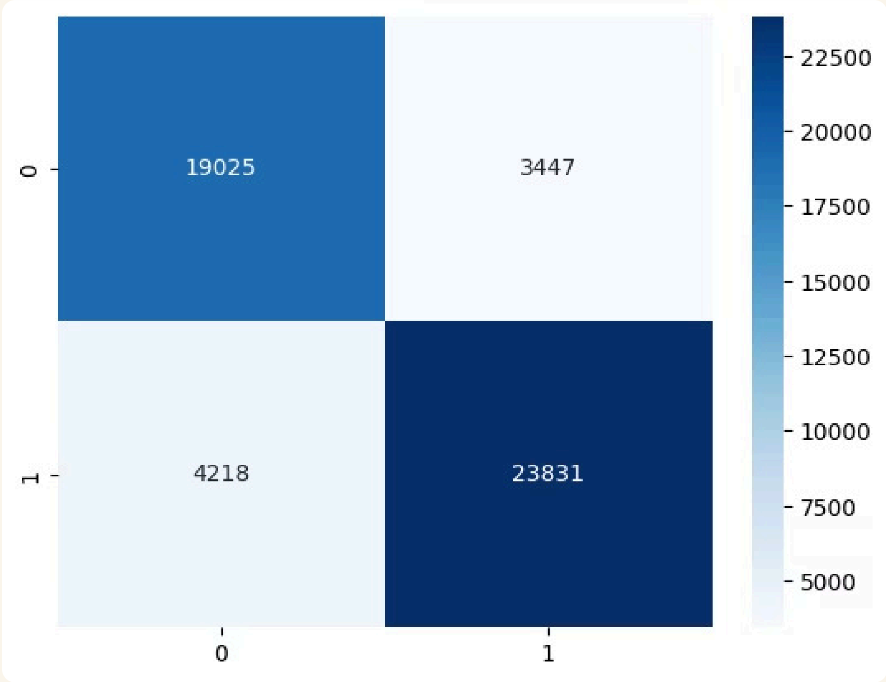
- **Targeted Feature Engineering:**
  - **The Action:** We created custom binary flags (e.g., `Critical_Support_Calls > 3`, `High_Risk_Age > 50`) based on our EDA findings.
  - **The Goal:** This gives simpler linear models (like Logistic Regression) a fair chance to "see" the sharp drops in customer behavior that they usually miss.
- **Data Transformation:**
  - **Encoding & Scaling:** Applied One-Hot Encoding for categorical features and Standard Scaling for numerical ones.
  - **The Goal:** Ensures all numbers are on the same playing field, so large values (like Total Spend) don't unfairly dominate the model's decisions.
- **Memory Optimization (Speed & Efficiency):**
  - **The Action:** Downcasted binary and encoded columns from default float types to `int8` (8-bit integers).
  - **The Goal:** Drastically reduces the dataset's size in memory, making model training much faster and preparing the app for a lightweight deployment.

Modeling — experiments, comparison & final selection

1st Logistic Regression Models :

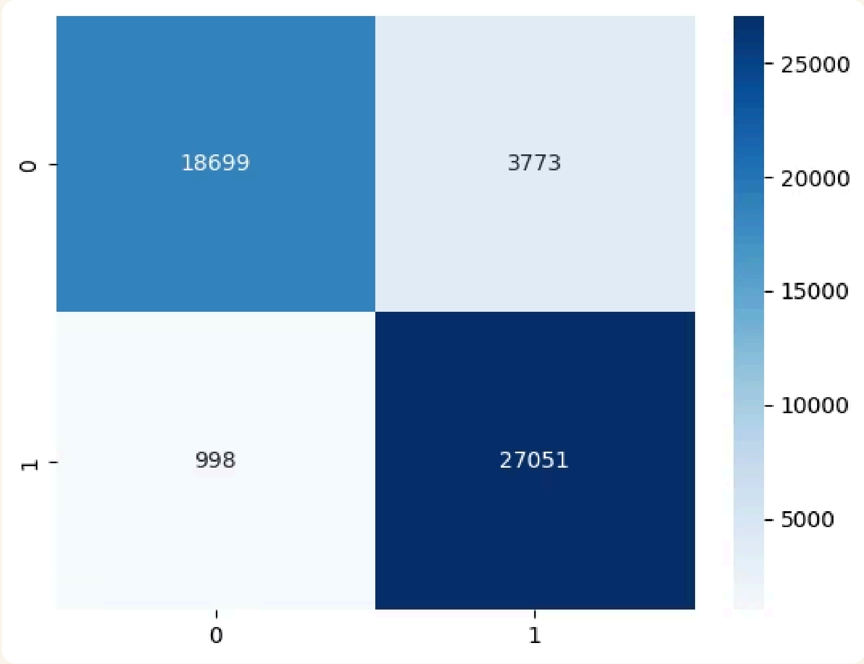
First Model On Original Features :

- **Baseline Performance:** Trained on original features. Achieved 85% Accuracy and 85% Recall.
- **The Problem:** The model missed a significant number of actual churners (False Negatives) because it couldn't "see" the sharp behavioral cliffs clearly.



Second Model On New Features :

- **Engineered Success:** Using our custom binary flags and shifting the decision threshold (to > 0.3) boosted Accuracy to 91%.
- **Business Value:** Recall jumped to 96%, drastically reducing False Negatives. The model is now much safer for retention campaigns.



2nd Random Forest Model :

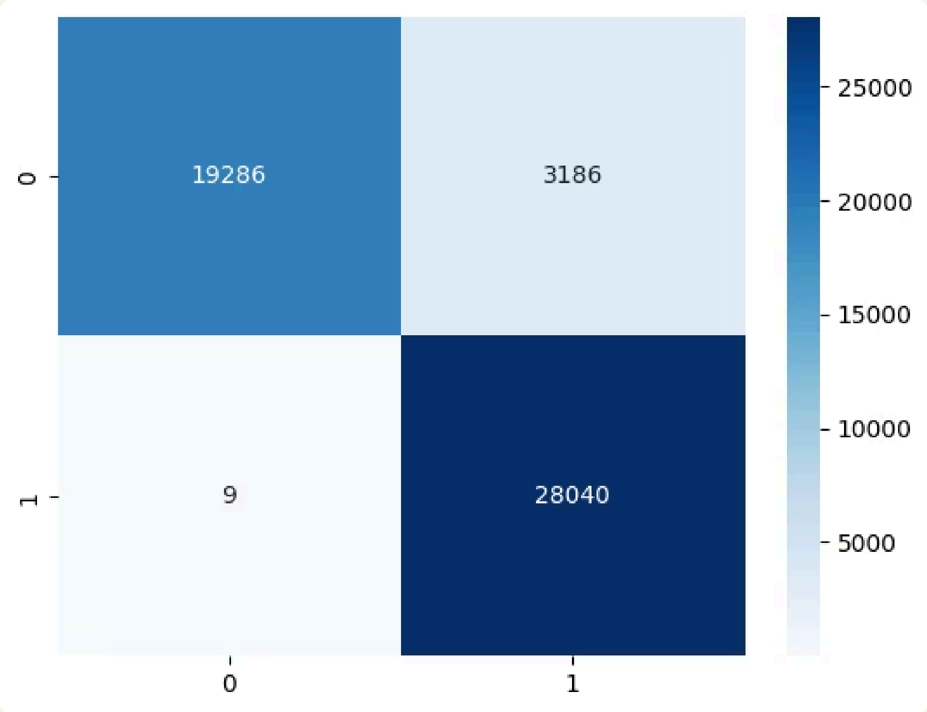
- **Performance Leap:** The model easily captured the complex behavioral "cliffs," jumping to ~94% Accuracy and ~93% F1-Score.

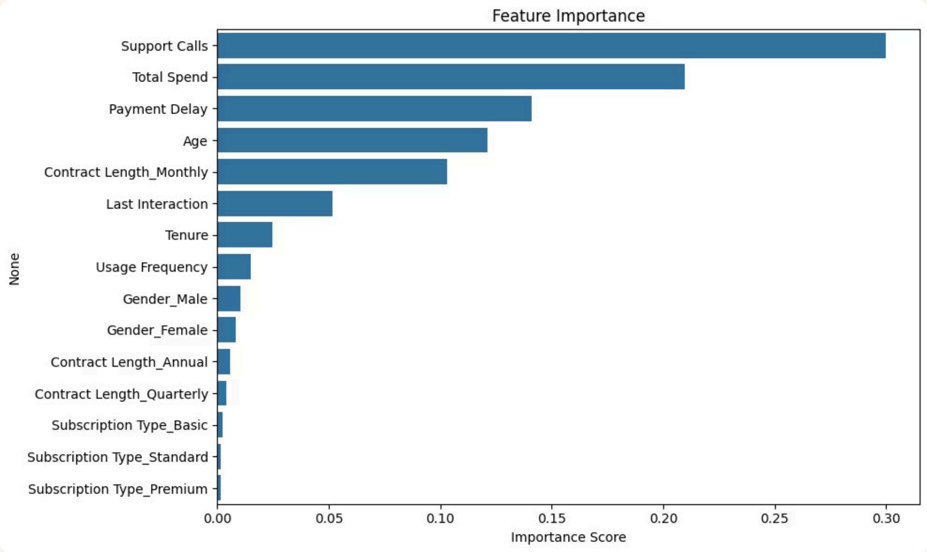
- **Exceptional Risk Detection:** The absolute standout metric is the Recall. The model reduced False Negatives to an unprecedented **9 cases** out of thousands.

- **The Business Safety Net:** In churn prediction, missing a churning customer is the most expensive mistake. Missing only 9 cases makes this model highly reliable for retention campaigns.

- **Validating Human EDA:** The model's internal logic perfectly mirrors our earlier visual analysis. It naturally selected Support Calls, Total Spend, and Payment Delay as the top three decision drivers.

- **Initial Deployment Choice:** Because of its unmatched ability to catch almost every at-risk customer, Random Forest was officially selected as our preliminary deployment model.

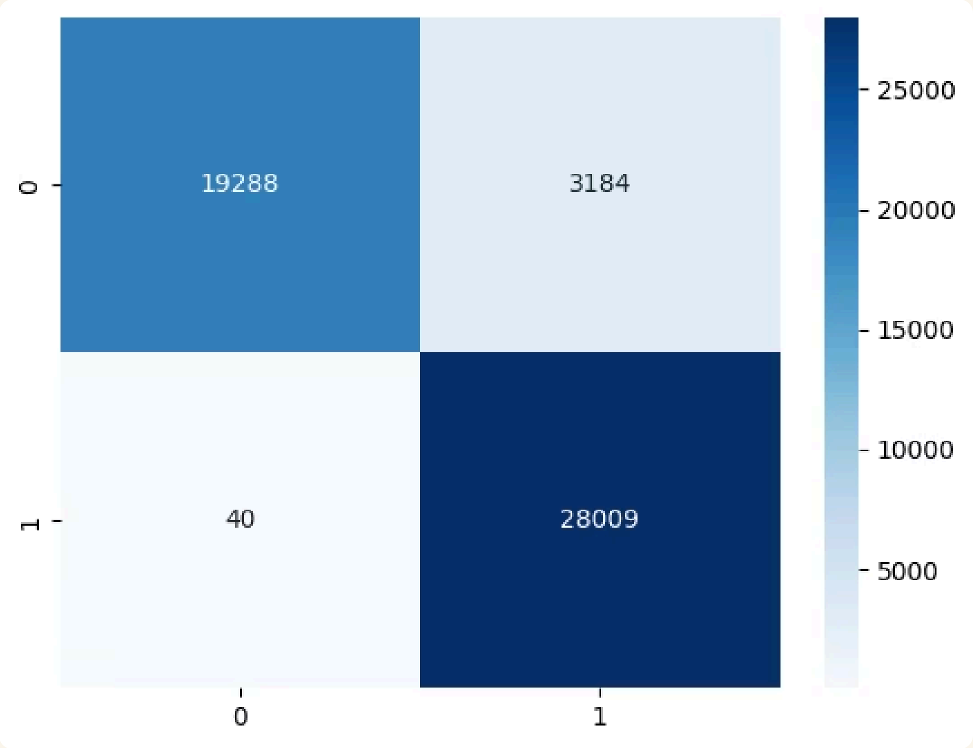


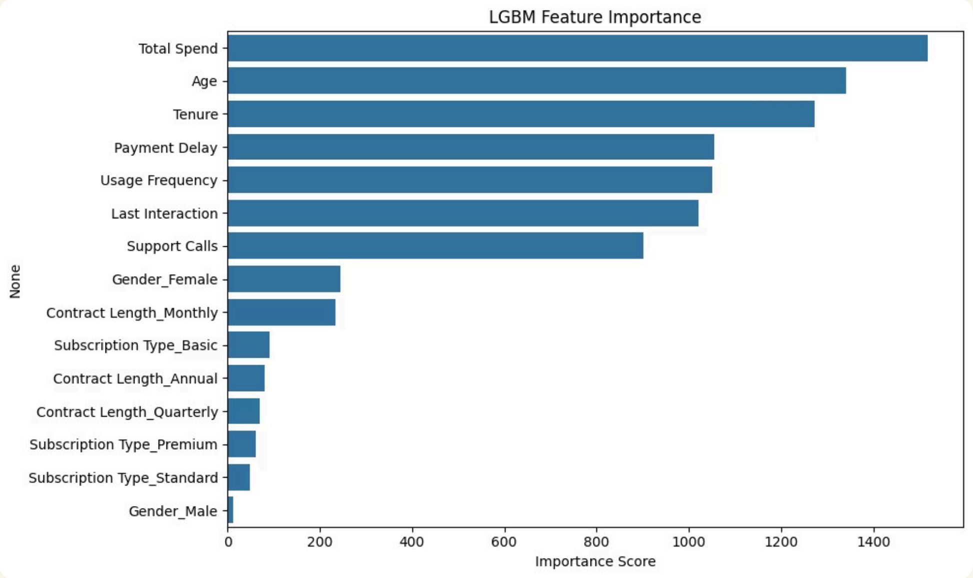


3rd Model LGBM Classifier :

First LGBM On Original Features :

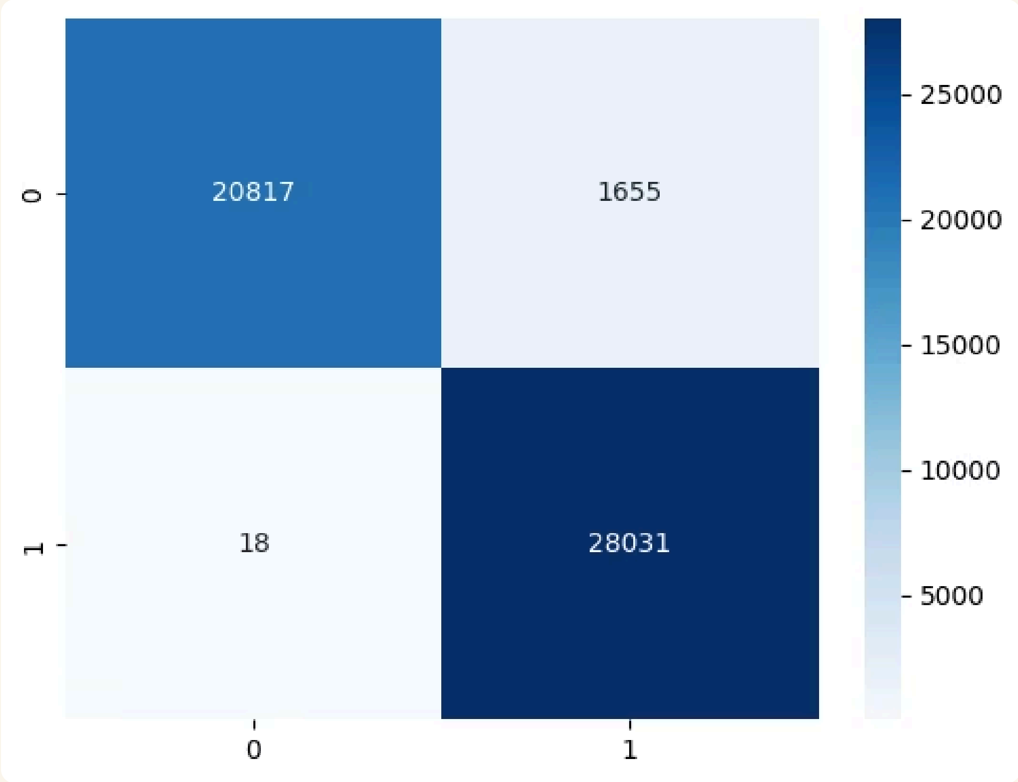
- **Solid, But Slightly Worse on Risk:** LightGBM matched Random Forest in general accuracy (~94%), but it was worse at our main goal: it missed more actual churners (40 False Negatives compared to RF's 9).
- **A Different "Brain" (The Surprise):** Looking under the hood, LightGBM's logic was totally different. While Random Forest focused heavily on Support Calls, LightGBM prioritized financial and demographic limits (Total Spend, Age, and Tenure).
- **The Setup for Optimization:** This unique way of thinking told us the model might be getting distracted by less important columns. This sparked our next idea: What if we filter out the noise and force it to focus?





Second LGBM On Selected Features :

- **The Filter (RFECV):** We applied Recursive Feature Elimination with Cross-Validation (RFECV) to strip away statistical noise and force the model to focus strictly on the most powerful predictive signals.
- **The Performance Leap:** The transformation was drastic. The model's overall Accuracy and F1-Score jumped to a massive **97%.**
- **Best of Both Worlds:** Most importantly, False Negatives plummeted from 40 down to just **18**. We successfully achieved the extreme risk-detection (Recall) of the Random Forest, but with the lightning-fast speed and tiny file size of LightGBM.



Models Performance Comparison (Class: Churn 1.0)

Model	Feature Set	Accuracy	Precision	Recall	F1-Score
Logistic Regression	Original Features	85%	87%	85%	85%
Logistic Regression	Engineered + Tuned	91%	88%	96%	92%
Random Forest	Original Features	94%	90%	**100%**	95%
LightGBM	Original Features	94%	90%	**100%**	95%
🏆 **LightGBM (Final)**	**Selected (RFECV)**	**97%**	**94%**	100%	97%

Thank You